

NETWORKS AND COMPLEXITY

Solution 14-9

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 14.9: A puzzle [Lvl. 4]

Determine which of the following Jacobians corresponds to a stable steady state (you won't need a computer):

$$\mathbf{J}_1 = \begin{pmatrix} -9 & -1 & 2 & 0 & 1 \\ 0 & -5 & 0 & 1 & 0 \\ 1 & 1 & -4 & 0 & 0 \\ 4 & 1 & 0 & -3 & 0 \\ 3 & 1 & 1 & 1 & -2 \end{pmatrix}, \quad \mathbf{J}_2 = \begin{pmatrix} -4 & 0 & 0 & 1 & 0 \\ 5 & 1 & 0 & 2 & 0 \\ 1 & -2 & -2 & 4 & 4 \\ 6 & 0 & 0 & -1 & 0 \\ -1 & -2 & 0 & 1 & -1 \end{pmatrix}$$

$$\mathbf{J}_3 = \begin{pmatrix} -1 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3 \\ -4 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 4 & 2 & 1 & -2 \end{pmatrix}, \quad \mathbf{J}_4 = \begin{pmatrix} -2 & 1 & 1 & -4 & 3 \\ 0 & -9 & 1 & 1 & 0 \\ 0 & 0 & -9 & 1 & 0 \\ 0 & 0 & 0 & -9 & 0 \\ 3 & -2 & 2 & 1 & -2 \end{pmatrix}$$

Solution

If we apply Gershgorin's theorem to the columns of $\mathbf{J}_1$ we see that all eigenvalues lie in a set of disks that are fully on the negative side of the complex plane, so all eigenvalues must be negative.

In $\mathbf{J}_2$, we can see that there must be an eigenvalue of -2 because -2 stands alone in the third column. When we then remove the third column and third row, it leaves -1 as the only element in the last column, so we can also remove the last column and the last row. Now we have a 1 that is alone in the second column, so there must be an eigenvalue of 1 and hence the system is unstable.

Matrix $\mathbf{J}_3$ is interesting. We can see there are positive and negative diagonal elements. So we won't be able to prove stability using Gershgorin's theorem. But we also are not able to prove instability using Gershgorin's theorem, because the elements in the row and column are large. Furthermore we cannot decompose the matrix into smaller matrices. But in this case we can make progress with our feedback rules. Note that the node that has the 1 as its diagonal element is only linked to node 5, which has a diagonal element of -2. Moreover the connection is via two positive elements so this is a positive feedback loop. If this feedback node were switched off we would have an eigenvalue of 1. If we then switch the feedback loop back on the positive feedback pushes the eigenvalue further away from the -2, so it becomes even more positive. Hence this is a Jacobian of an unstable state.

In $\mathbf{J}_4$, the variables 2, 3, and 4 decouple from variables 1 and 5. So we can treat these parts separately. Variables 2,3,4 have three eigenvalues of -9, which is evident from the triangular form of the central block (we can isolate one -9 then the next, and finally the third one). Once the central variables are removed, we are left with a much smaller matrix for variables 1 and 5,

$$\begin{pmatrix} -2 & 3 \\ 3 & -2 \end{pmatrix} \quad (1)$$

The corresponding eigenvalue problem can be easily solved in various ways. For example the characteristic polynomial is very easy to factorize for this matrix. Alternatively, note that the two variables are only coupled by a positive feedback loop, so the eigenvalues are real. Moreover, the determinant is $4 - 9 = -5$. Since the determinant is the product of the eigenvalues, we might ask how the product of two real eigenvalues can give a negative result. Clearly, one eigenvalue must be positive, and one negative, which proves that this is an unstable case.

Alternatively, we could have guessed the eigenvectors based on the symmetry of this matrix, but we will leave that for a later chapter.